

Stored Procedures

What are Stored Procedures?

Stored Procedures are database functions used to execute complex SQL logic in a procedural way.

Stored procedures in Redshift allow procedural programming using PL/pgSQL, making it possible to automate complex SQL workflows with variables, conditions, loops, DDL, and DML operations directly inside the cluster.

They help perform:

- Conditional logic
- Loops and iterations
- Variable handling
- Multiple SQL operations together

Supported Operations

DDL Commands

Used for structure changes:

```
CREATE  
ALTER  
DROP
```

DML Commands

Used for data operations:

```
SELECT  
INSERT  
UPDATE  
DELETE
```

Why Stored Procedures?

Normal SQL is not procedural.

Stored procedures allow:

- Step-by-step execution
- Better automation
- Reduced manual query execution
- Complex database workflows

Procedural Language Used

Amazon Redshift uses: `PL/pgSQL` (PostgreSQL Procedural Language)

It is similar to: `Oracle PL/SQL`

So users familiar with Oracle procedures can easily understand Redshift procedures.

Advantages of Stored Procedures

Runs Inside the Cluster

Procedures execute directly inside Redshift.

Benefit

- Reduces remote communication
 - Improves execution efficiency
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Important Commands

Create Procedure

Used to create a stored procedure.

```
CREATE PROCEDURE
```

Execute Procedure

Used to run the procedure.

```
CALL procedure_name( );
```

Limitations

Maximum source code size: 2MB

Maximum parameters: 32

Typical Use Cases

Stored procedures are commonly used for:

- ETL workflows
 - Batch processing
 - Data validation
 - Automated database tasks
 - Complex business logic
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